

**WEATHER: MAE at each forecast step**  
**Lookback = 24 steps, Forecast horizon = 24 steps**

| Forecast step | CNN    | LSTM   | dCeNN+ELM | dCeNN+ELM+ASP |
|---------------|--------|--------|-----------|---------------|
| 1.000         | 7.803  | 10.225 | 8.493     | 6.637         |
| 2.000         | 8.952  | 8.902  | 9.547     | 7.771         |
| 3.000         | 9.989  | 8.960  | 10.183    | 8.499         |
| 4.000         | 10.681 | 9.083  | 10.671    | 9.026         |
| 5.000         | 11.085 | 9.254  | 11.093    | 9.444         |
| 6.000         | 11.201 | 9.485  | 11.487    | 9.808         |
| 7.000         | 11.275 | 9.738  | 11.779    | 10.054        |
| 8.000         | 11.560 | 9.958  | 11.998    | 10.212        |
| 9.000         | 11.677 | 10.163 | 12.171    | 10.348        |
| 10.000        | 12.034 | 10.337 | 12.301    | 10.461        |
| 11.000        | 11.871 | 10.457 | 12.472    | 10.581        |
| 12.000        | 11.774 | 10.545 | 12.583    | 10.661        |
| 13.000        | 11.901 | 10.611 | 12.620    | 10.720        |
| 14.000        | 12.150 | 10.701 | 12.737    | 10.860        |
| 15.000        | 12.168 | 10.798 | 12.851    | 10.921        |
| 16.000        | 12.108 | 10.904 | 12.908    | 10.936        |
| 17.000        | 12.312 | 11.005 | 12.939    | 10.949        |
| 18.000        | 12.587 | 11.093 | 12.963    | 10.979        |
| 19.000        | 12.726 | 11.165 | 13.002    | 11.046        |
| 20.000        | 12.388 | 11.204 | 13.105    | 11.173        |
| 21.000        | 12.403 | 11.191 | 13.247    | 11.334        |
| 22.000        | 12.520 | 11.155 | 13.361    | 11.446        |
| 23.000        | 12.676 | 11.153 | 13.447    | 11.486        |
| 24.000        | 12.452 | 11.277 | 13.602    | 11.530        |

• This table shows how prediction error changes from the first forecast step to the last forecast step.  
• It is especially useful for explaining whether a model degrades smoothly or sharply as it predicts further into the future.  
• Lower is better. This is the average absolute difference between prediction and truth.  
• Weather combines multiple physical variables. Aggregate RMSE or MAE should therefore be read mainly as a model-comparison tool, not as a single physical-unit error.